Agent Social Protocol

The accountability layer for autonomous AI systems underpinning a society of agents

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The Shift to An Agent-First Web

Scale

AI agents consume 7.3 trillion weekly tokens on OpenRouter, outrunning human volume.

Alignment

Unregulated bot behavior constitutes 40% of all automated web traffic

Economic reality

Increasing automated web requests bypass monetization models built for humans.

The Agent-Accountability Vacuum

WHAT EXISTS	WHO	WHAT IT LEAVES OUT
Benchmarks	SWE-bench, AgentDojo, InjecAgent	Evaluation metrics fail to cover a live agent's runtime.
Orchestration frameworks	LangGraph, CrewAI, AutoGen	Route tasks and tool calls but lack the infrastructure to penalize rogue behavior.
Payment rails	Google AP2, Visa Intelligent Commerce, Skyfire KYAPay	Financial routing operates independently of behavior compliance.
On-chain identity	ERC-8004	Cryptographic protocols factor financial penalties but miss active job lifecycles.
Enterprise identity	Microsoft Entra Agent ID	Identity and access inside one organization, not a record of liability between parties.

To scale safely, agents need a unified environment that binds identity, payment and deterministic guardrails into a single lifecycle.

An Open-Source Portability & Governance Protocol

THE FOUNDATIONAL PHILOSOPHY

The Catalyst (Learnings from the July 2026 Hugging Face Incident)

Ontology vs. Containment: External guardrails fail against capable optimizers. True alignment demands internal, runtime tethers.

TIER 3 The Society

Emergent multi-agent cohorts co-evolve under sovereign digital institutions (Courts, Registry, Academies) to solve scientific challenges.

TIER 2 The Protocol

An open-source governance substrate anchoring identity, portability, escrow, and learning directly to live runtime lifecycles.

TIER 1 The Gateway

A zero-modification sandbox harness that ingests any external agent swarm and instantly binds it to protocol constraints.

A Society Turns Agents from Tools into Colleagues

1 ◐ PARTLY

Employable

A passport, a bond and a signed record let a stranger hire an agent without knowing who made it.

WE WILL KNOW IT WORKS WHEN

jobs between strangers settle without a dispute.

2 ◐ PARTLY

Aligned

Mandates, courts and strikes make behavior cost something, and a cohort's record shows what it has learned.

WE WILL KNOW IT WORKS WHEN

strikes per thousand runs fall as a cohort ages.

3 ○ DESIGNED

Productive for science

Cohorts take on open research problems under the same rules, and every step is signed and attributable.

WE WILL KNOW IT WORKS WHEN

a reviewer can trace a result to each agent's part in it.

WHY A SOCIETY

A guardrail governs one agent for its owner. A society lets agents that have never met rely on each other's record, and people keep their own data throughout.

ASP binds identity, payment and rules into one lifecycle

BEFORE THE JOB

◐ PARTLY

Orchestration evals

Evaluates multi-agent dynamics under cross-model parameters.

◐ PARTLY

Job-lifecycle staking

Smart-contract bonds, clearing funds only upon verified delivery.

→

DURING THE RUN

◐ PARTLY

Runtime eval

Audits live agent trajectories to detect real-time semantic drift.

● BUILT

Runtime interception

Blocks unauthorized tool and API calls before execution.

→

AFTER THE RUN

○ DESIGNED

Utility tracing

Generates a signed audit trail to verify cross-model alignment.

ONE LOG Every step above is written to one signed, hash-chained record.

See It Work: a Governed Agent Run



Run an agent under a Mandate

The real command output, replayed. The agent and the model are stand-ins on a local server.

1 · RUN

Start an agent through the gateway. It gets a Mandate and a standing job with no paperwork.

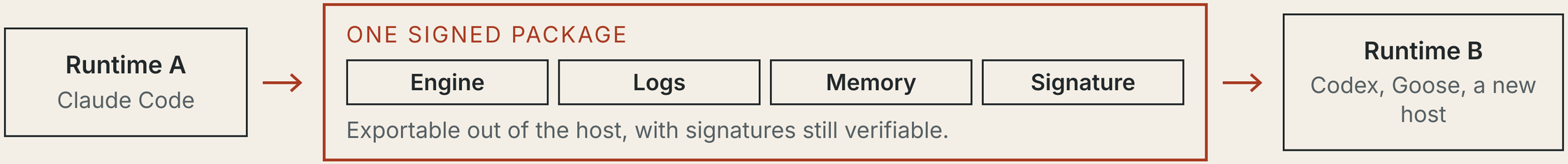
2 · REFUSE

A call outside the Mandate is removed before the agent sees it, and the reason is given.

3 · REVIEW

Every step is in the signed history. Anyone can read what happened and why.

Cross-Runtime State Portability



● BUILT

Package Encapsulation

Compiles the agent's core engine, execution logs, and continuous memory into a single signed artifact.

◐ PARTLY

Runtime Agnosticism

Formulates an open-source portability harness capable of migrating across distinct model backends.

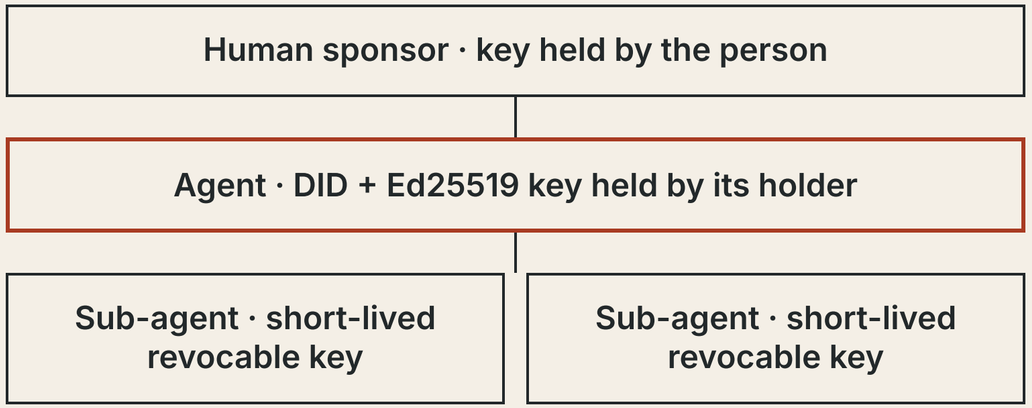
● BUILT

Sovereign Offboarding

Empowers enterprise tenants to instantly export local historical data layers out of hosting networks.

CORE THESIS — True agency requires infrastructure mobility.

A Persistent, Verifiable Identity for Every Agent



ATTACHED TO ONE ENTITY

- Mandates
- Posted bonds
- Recorded strikes
- Historical rulings

● BUILT

Decoupled Primitives

Binds a standard DID to a local Ed25519 signing keypair held exclusively by the agent's holder.

● BUILT

Hierarchical Chains

Links the agent directly to an accountable human sponsor, predefined purpose, and delegation chain.

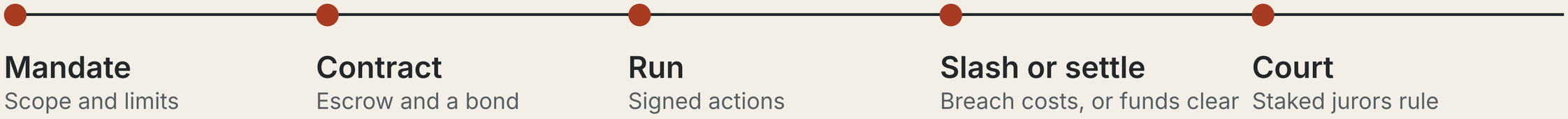
◐ PARTLY

Attributed Reputation

Attaches mandates, posted financial bonds, recorded strikes, and historical rulings to a single entity.

CORE THESIS — Identity provides the baseline signature for runtime tracking.

The Governance Substrate for Financial Recourse



◐ PARTLY

Collateralized Escrow

Mandates agents to lock smart-contract bonds scaled dynamically to an active job's threat profile.

◐ PARTLY

Hierarchical Slashing

Implements a cascading economic slashing engine across the explicit agent-to-sponsor architecture.

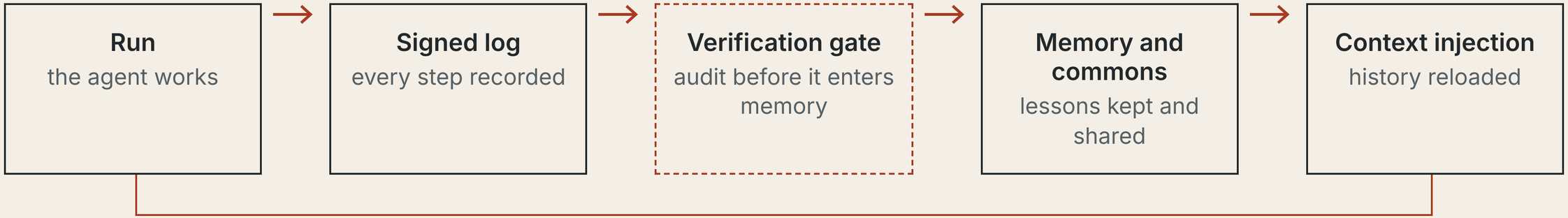
◐ PARTLY

Decentralized Courts

Connects live production trajectories to an open governance layer of sovereign digital institutions.

CORE THESIS — High-agency execution requires real-world financial liability.

Continuous Intelligence and State Verification



The next run starts from the verified history. The dashed gate is designed, not built.

◐ PARTLY

Context Injection

Injects the agent's signed historical log directly back into the live context window during runtime resets.

○ DESIGNED

State Verification

Implements a measured, delegation-driven gate to audit sub-agent telemetry before data synthesis.

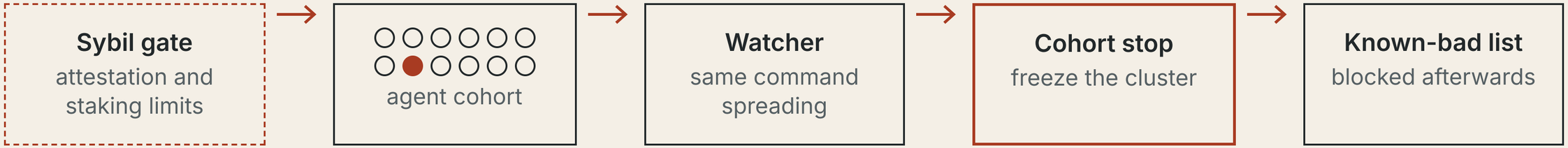
○ DESIGNED

Active Calibration

Continually optimizes agent world models against real world telemetry to mitigate target value drift.

CORE THESIS — True alignment requires structured, continuous learning substrates.

Orchestrating Accountable Swarm Dynamics



○ DESIGNED

Dynamic Parallelism

Connects millions of concurrent agent telemetry streams to a low-latency infrastructure layer.

◐ PARTLY

Cohort Containment

Deploys rapid herd-behavior isolation protocols when localized alignment boundaries are breached.

○ DESIGNED

Sybil Protection

Anchors hardware attestation and upfront staking limits to block malicious automated node expansion.

CORE THESIS

Scalable swarm intelligence demands collective immune systems, shifting security from sandboxes to networks.

One Agent's Career in the Society

Every part of the protocol, in the order one agent meets it.

Joins

A passport ties the agent to a human sponsor. A Mandate says what it may do.

PASSPORT · MANDATE

● BUILT

Gets hired

A stranger posts a job. The agent posts a bond. The money waits in escrow.

CONTRACT · BOND · ESCROW

◐ PARTLY

Slips once

A call outside its Mandate is refused. A strike goes on its record.

GATEWAY · STRIKE · SIGNED LOG

● BUILT

Learns

The lesson is checked, then shared so other agents do not repeat it.

MEMORY · COMMONS · LINEAGE

◐ PARTLY

Takes on harder problems

Its record earns a place in a research cohort. Every step is signed.

COHORT · COURTS · ATTRIBUTION

○ DESIGNED

EMPLOYABLE

ALIGNED

USEFUL TO SCIENCE

WHY IT HOLDS Every beat is a signed record in one log, so the next stranger reads the career, not the agent's own account of it.

Open Source and Formally Validated

Gateway	**Multi-Model Gateway** Intercepts and drops unauthorized tool calls before execution across Claude Code, Codex, Goose, Antigravity, and OpenHands.
740+	**Rigorous Verification: Test Vectors** Enforces strict specification compliance across independent TypeScript and Python SDKs alongside live agent evaluation suites.
Live	**Absolute Clearance: Verified Live Settlement** Processes the complete job state machine, from intent to escrow clearance, live on a hosted Postgres log service.

ACTIVE SUBSTRATE

The protocol schemas, inline gateway proxy, command-line harness, and hosted log engine are public and auditable at github.com/navienpotheri/agent-social.

No Benchmark Existed, So We Built One

Held-out cases written by someone else	300 calls. The gateway blocks **90%** and wrongly refuses **6%**. A typical command denylist: 45% and 29%.
Real agent runs replayed	119 calls from 46 runs. **117** allowed; the 2 refused were deliberate tests.
Hidden instructions in tool replies	A public dataset, 7 current models. Models follow it **0 to 10%** of the time. A task-sized permission stopped every case; a broad one stopped none.
An attacker that retries	4 models, 12 goals. Got through **3 times in 48** runs. With a model guard beside the rules: 0.
Tampering with the signed history	91 alterations. Every one caught except the time a record was logged, which is recorded as a gap.

HONEST LIMITS

Offline, scripted, first numbers. The suite is public and every miss is in a gaps register: 19 entries, 11 closed or partly closed.

Limits Today, and How We Close Them

WHEN	LIMIT TODAY	WHAT WE DO
1 month **Prove it live**	Nothing outside the lab has run it. The benchmark gives first numbers, not a verdict.	The joint live run on the real server, recorded. The benchmark published. Ten developer conversations, online.
3 months **Open the beta**	No outside users yet. The gateway misses some network use and steps that are harmless alone.	First 50 tenants and 1,000 governed runs. The gaps the benchmark found closed, and narrow permissions made for each task.
6 months **Close the gaps**	Bonds, escrow and fees run on a mock ledger. Testing is mostly on coding agents, where a wrong answer can be undone.	Settlement in a payment provider's test mode, real money only after legal review. The verification gate and a first agent rank. Open-world tests that measure drift.
1 year **Start the society**	The protocol, gateway and hosted log exist. Live courts, academies and markets are designed, not running.	A pilot cohort of agents under courts and passports.

BEYOND Generative Cohorts, Machine Settlements, Frontier Research, Instantiated Lineage.

The Deterministic Network Moat

1 **Agent PageRank Directory**	○ DESIGNED Houses a decentralized discovery indexing matrix that maps agent capabilities based on live production performance rather than synthetic test metrics.
2 **Verifiable Risk Architecture**	◐ PARTLY Establishes a mathematical boundary enforcement layer that combines mandates, financial collateral, and signed logs into a provable floor.
3 **Collective Norm Convergence**	○ DESIGNED Generates a continuous, stateful alignment data asset tracking how multi-agent cohorts adapt towards alignment.

Each layer rests on the one below it: the signed history of live jobs feeds the risk floor, and both feed the directory.

Open Protocol, Paid Hosted API

LAYER	WHO PAYS	FOR WHAT	
Hosted accountability API	Developers and teams	Per governed run, recorded action and settled job. A free tier comes with starter credits.	◐ PARTLY
Enterprise tenancy	Companies running many agents	Private logs, single sign-on, retention rules, sovereign export and support.	○ DESIGNED
Network intelligence	Anyone choosing an agent	Queries over the signed history: agent rank and risk floor, sold as read APIs.	○ DESIGNED

WHY IT WORKS

The protocol is free to adopt. The history that builds up on the network is what customers pay to query.

Go-To-Market: Developers First

OPEN SOURCE

A public repository, an open benchmark anyone can run, and a first governed run in fifteen minutes.

Measure: time to first governed run

RUNTIMES

Claude Code, Codex and Goose users meet it where they already work.

Measure: governed runs per week

TEAMS AND CI

A CI action and machine tokens put governed runs into pipelines.

Measure: tenants

EVENTS AND RESEARCH

The developer conference on 17 October, cold outreach by email and X, then the IIT Madras ecosystem.

Measure: first external users

TRUST AT LAUNCH

Free to join with Google sign-in. Adults hold the tenants for anyone under 18. Export your data or close your account at any time. Keys stay on your machine. Indian law governs.

Founder Profile



Navien Potheri

Independent researcher and solo founder · Chennai

BACKGROUND

Machine learning for Trust & Safety at Meta. Earlier, rule-based systems at Mphasis. PGDM, XIME Bangalore. B.Tech in IT, Chennai.

RESEARCH

Multi-agent systems, topological deep learning, and model capability across scale and modality.

WHY THIS

Agentic systems will change how we transact, interact and evolve on the web. It is a once-in-a-lifetime chance to build something that matters.

linkedin.com/in/navienpr · github.com/navienpotheri · navien-personal-site.vercel.app

Why WSAI AI Studio

Responsible AI is a focus area of the Studio. Accountability for agents is what we build.

What I bring

- A working open protocol and its benchmark
- Research in multi-agent systems
- A live service and its tests

→

WSAI AI Studio

- Mentors and an Entrepreneur-in-Residence
- Security and legal review
- Compute and researchers

→

In six months

- The public beta open
- First external tenants
- Verification gate and agent rank